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BIG MART SALES USING SUPERVISED MACHINE LEARNING ALGORITHMS

ABSTRACT

This paper focuses on developing a sales prediction model for Big Mart, a supermarket chain, using machine learning algorithms. The developed model aims to provide Big Mart with accurate sales forecasts, enabling better decision-making, improved profitability, and enhanced customer service. In today's data-driven world, accurate sales forecasting plays a critical role in decision-making processes for retail businesses. This project focuses on predicting sales of various products at Big Mart stores using supervised machine learning algorithms. The dataset includes historical sales data, store-related information, and product-level features such as item type, outlet size, location, and visibility. The goal is to build predictive models that can estimate future sales based on these variables.

Several supervised learning algorithms, including Linear Regression, Decision Trees, Random Forests, and XGBoost, are implemented and evaluated for their performance. Data pre-processing techniques such as handling missing values, encoding categorical variables, and feature scaling are applied to enhance model accuracy. Model performance is assessed using metrics such as Root Mean Squared Error (RMSE) and R-squared score.

The project demonstrates that ensemble methods like Random Forest and XGBoost provide better predictive accuracy due to their ability to handle non-linear relationships and complex interactions among features. The results can aid Big Mart in inventory management, targeted marketing, and strategic planning, ultimately leading to increased profitability and customer satisfaction.

INTERNAL GUIDE

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